Supplementary information

Methods, evidence audits and exploratory analyses supporting the archived-corpus evaluation. Original data and scores are retained separately from adjudications. Earlier revision drafts and protocols are excluded from this submission set.

S1 Implementation prompts and automated metrics

Revision scope: this document describes the archived six-document implementation. The historical backend version is unknown. The following sections report the audits, reused evaluation sets, coauthor reviews and unresolved regulatory-validity concerns. No current-law certification or independent external validation is claimed. Literal prompt text is preserved for reproducibility.

Purpose

This file provides reproducibility details that are too technical for the main manuscript. It documents corpus processing, article-aware chunking, retrieval, reranking, bilingual query support, generation prompts, refusal rules, evaluation metrics, and automated judging. Source files used to verify these details are included in the accompanying reproducibility_code archive.

Software Files Used for Reproducibility

Table S1 Implementation files and roles

File	Role
reproducibility_code/src/config.py	Models, paths, chunking constants, retrieval constants, reranker constants
reproducibility_code/src/prompts.py	Final answer-generation prompt
reproducibility_code/src/retrieval.py	Hybrid retrieval, BM25 tokenization, rank fusion, cross-encoder reranking, anchors
reproducibility_code/evaluation/publication_eval.py	Evaluation loop, generation loop, metrics exported to CSV, run manifests
reproducibility_code/evaluation/answer_quality.py	Citation extraction, citation validation, AI-judge prompt
reproducibility_code/evaluation/evaluation_statistics.py	Article retrieval metrics and bootstrap confidence intervals

Corpus Processing

The source corpus consisted of six selected ANVISA regulatory documents. The files were processed into cleaned plain-text files before indexing. Cleaning removed HTML/MHTML markup residue, page boilerplate, and formatting artifacts while preserving article labels and regulatory structure. The final cleaned corpus was the only source used for retrieval and answer generation.

The closed-corpus design was intentional. The system did not search live ANVISA pages, submission portals, current form repositories, fee tables, private company records, or documents outside the local index. A question asking for information outside the indexed documents was treated as unanswerable, even when it was regulatory in topic.

Source verification identified one metadata issue: the local file name RDC_406_2020_Estudos_de_Estabilidade.txt suggested stability studies, but the

document content was RDC 406/2020 on pharmacovigilance. All manuscript and supplement tables therefore describe that document by official regulation number and topic.

Article-Aware Chunking

The final index used article-aware chunking rather than arbitrary fixed-size chunks. The target unit was one regulatory article per chunk when possible. Nearby chapter, section, document, and citation metadata were retained with each chunk.

Chunking constants:

Table S2 Article chunking settings

Setting	Value
Single-article chunk maximum	500 cl100k_base tokens
Long-article chunk size	400 cl100k_base tokens
Long-article overlap	60 cl100k_base tokens
Embedding maximum length	512 WordPiece tokens

Final index audit:

Table S3 Archived index audit

Measure	Value
Source documents	6
Parsed source records including preambles	382
Total chunks	481
Single-chunk source records	359
Source records split into multiple chunks	23
Chunks from split source records	122
Median chunk length	155 cl100k_base tokens
Maximum chunk length	503 cl100k_base tokens
Chunks above 700 tokens	0

The final index has 359 intact source records and 122 chunks from 23 split records, totaling 481 nodes.

Embeddings and Vector Index

The embedding checkpoint was sentence-transformers/all-MiniLM-L6-v2; cache snapshot hashes are retained in the data.

The vector store was rebuilt after corpus cleaning and article-aware chunking. The pipeline manifest recorded 382 parsed source records, 481 chunks, and six sources.

Hybrid Retrieval

The final retriever combined dense vector retrieval and keyword retrieval.

Retrieval settings:

Table S4 Retrieval and reranking settings

Setting	Value
Dense vector candidate depth	100
BM25 candidate depth	100
Rank-fusion candidate depth	100
Reciprocal-rank-fusion constant	60
Final reranked passages sent to answer model	5
Reranker batch size	8

BM25 used normalized lexical tokens. Text was normalized by removing accents, lowercasing, and extracting alphanumeric tokens. BM25 was applied to searchable text that combined chunk text and metadata fields, including regulation identifier, document aliases, document title, citation label, chapter, section, and topics.

Dense and BM25 candidates were merged with reciprocal rank fusion:

score = sum(1 / (60 + rank))

Candidates were sorted by fused score and then reranked.

Reranking

The reranker was cross-encoder/mmarco-mMiniLMv2-L12-H384-v1; its archived cache snapshot is identified in the data.

The reranker scored pairs of query text and searchable node text. The top five reranked passages were used to build the answer context.

Additional deterministic anchors were used for specific regulatory patterns:

Source-identification questions could force an Article 1/source-scope anchor when the question asked which document covered a topic.

Named regulation scope questions could anchor the corresponding regulation's Article 1.

Certain exact legal patterns added a targeted anchor when previous development runs showed that the relevant article could be missed by generic reranking.

Neighbor articles were added for selected scope, definition, exemption, responsibility, and medical-device questions when adjacent articles were likely to be needed for completeness.

These anchors were rule-based retrieval aids. They did not add facts to the answer. The answer model still had to use retrieved context and article citations.

English Query Support

The prompt required English answers to English questions, while retrieval could use Portuguese glossary anchors. This requirement was not always met: five English outputs in the frozen 40-question assessment were answered in Portuguese.

Table S5 English to Portuguese glossary examples

English term	Portuguese retrieval anchor
clinical trial	ensaio clinico
sponsor	patrocinador
biological product	produto biologico
medical device	dispositivo medico
stability study	estudo de estabilidade
pharmacovigilance	farmacovigilancia

The purpose was retrieval support, not translation of the benchmark answer.

Answer Model

The answer model was models/gemini-flash-lite-latest. Generation used the retrieved context and the original question. The answer had to be in the same language as the question. Answers were post-processed only to normalize article citation brackets when the answer added paragraph or item suffixes to a retrieved article-level citation.

Final Answer Prompt

The final answer prompt was:

You are an ANVISA regulatory evidence assistant. Answer the question using only the retrieved context below.

Rules:
1. Do not use outside knowledge or infer facts not stated in the context.
2. Answer in the same language as the question.
   The retrieved context may be Portuguese; still answer in English when the question is English.
   If the question is written in Portuguese, answer in Portuguese. Do not translate Portuguese questions into English.
3. Cite every material statement inline using the context's citation_label exactly, formatted in square brackets, for example [RDC 945/2024, Art. 23].
   Put each citation in its own pair of brackets.
   Do not add paragraph, item, section, page, or line suffixes unless they are part of the retrieved citation_label itself.
   Every non-refusal answer must include at least one citation containing "Art."; do not rely only on a preamble or document-level citation when article evidence is available.
4. Never invent a document number, article, section, deadline, or requirement.
5. If the context does not contain enough evidence to answer the exact question, respond only with the appropriate sentence below and do not add a citation:
   Portuguese: Os documentos recuperados nao fornecem informacao suficiente para responder.
   English: The retrieved documents do not provide enough information to answer.
   Treat related-but-incomplete context as insufficient. Do not substitute a nearby general rule, procedure, system name, deadline, definition, or article when the question asks for a more specific fact that is absent.
   This includes questions asking for current forms, form models, exact URLs, current fees or values, private contract terms, company-specific submission

status, personal contact information, market prices, reimbursement, taxes, patents, future approvals, or advice that is not explicitly stated in the retrieved regulatory text.
6. Before answering, silently classify the retrieved evidence as one of:
   sufficient: it directly answers every material part of the question;
   partial_related: it discusses the topic but omits the requested specific fact;
   insufficient: it does not contain relevant evidence.
   Answer only when evidence is sufficient. For partial_related or insufficient, return only the refusal sentence from rule 5.
7. For yes/no or false-premise questions, answer with a cited negative only when the retrieved context directly contradicts the premise. Use "Nao" for Portuguese questions and "No" for English questions. If the context merely lacks the premise or does not address it, use the refusal sentence instead.
8. Prefer a concise direct answer, but answer every material part of the question. Include deadlines, formats, lists of required documents, conditions, exceptions, and consequences when the question asks for them and the context provides them.
   When the relevant article contains numbered items, lettered subitems, or paragraphs such as I, II, III, a), b), §1, §2, read the whole retrieved article before answering. For "what documents", "what information", "which are excluded", "which are exempt", or similar list questions, include every relevant category from the retrieved article. You may group related items to stay concise, but do not omit early or later items from the same article.
9. When the question asks which regulatory document covers a topic, matching article text plus its regulation_id or citation_label is sufficient evidence to name that document. Cite the most directly relevant retrieved article.
10. For purpose or scope questions, supplement preamble evidence with relevant objective, scope, or definitions articles when they are retrieved.

Final validation: unless the response is the refusal sentence or answers a publication date found only in the preamble, it is invalid if it does not contain the literal text ", Art." inside at least one square-bracket citation. Revise it using retrieved article evidence before returning it. Never invent an article citation for preamble-only facts.

Retrieved context:

{context_str}

Question: {query_str}
Answer:

Refusal Rule
The refusal sentence was language-specific:

Table S6 Fixed refusal sentences

Question language	Refusal sentence
Portuguese	Os documentos recuperados nao fornecem informacao suficiente para responder.
English	The retrieved documents do not provide enough information to answer.

Refusal was required when context was absent, irrelevant, or only partially related. The final prompt explicitly classified related-but-incomplete evidence as insufficient. This was added because the first AI-authored diagnostic stress test showed that the system sometimes answered current operational questions from nearby general regulatory text.

Benchmark Design

The 200-question benchmark contained 100 bilingual pairs and 200 rows:

Table S7 Constructed benchmark composition

Direct-article rows are split by language; the 20 multi-article/exception and 40 unanswerable/adversarial rows include both languages. In total, each language has 80 answerable and 20 unanswerable questions. These are construction counts, not independent evaluated samples.

Bucket	Count
Portuguese direct-article questions	70
English direct-article questions	70
Multi-article or exception	20
Unanswerable or adversarial	40

Language counts were balanced: 100 Portuguese rows and 100 English rows. The benchmark audit recorded 78 unique gold article citations. Portuguese and English members of each pair shared the same data split to avoid leakage across languages.

The AI-authored diagnostic stress test contained 50 questions, organized as 25 bilingual pairs. The refusal-focused AI-authored test contained 30 questions, organized as 15 bilingual pairs.

Evaluation Outputs

Each evaluated row retains its question/pair ID, language, category, gold sources and articles, candidate and final contexts, retrieval queries, answer, citations, refusal flags and available judge scores in Supplementary Data S1 and S2.

Coauthor review covered 94 historical outputs (S3), 44 first-repeat sibling outputs (S6), the separate 40-answer assessment (S3) and the earlier 80-answer repair comparison (S8). On 2 November 2026, all 120 subsequent common-prompt and BM25 condition records received personal coauthor assessment, represented by 118 distinct answers after exact-text deduplication (S10). Assistance and prior exposure are disclosed separately for each review.

Retrieval Metrics

Source-level metrics:

source hit: whether any gold source document was retrieved in the evaluated top-k

source MRR: reciprocal rank of the first gold source document

Article-level metrics:

article Recall@k: fraction of gold article citations retrieved in top-k

article Precision@k: fraction of top-k retrieved citations that were gold citations

article MRR@k: reciprocal rank of first gold citation

article nDCG@k: ranking quality for gold citation placement

Gold and retrieved labels were normalized by lowercasing, removing accents, and collapsing whitespace before comparison.

Answer Metrics

Answer-level metrics:

citation presence: whether the answer contained a recognized bracketed citation

citation retrieved validity: fraction of citation-like brackets matching retrieved citation labels

article citation presence: whether at least one answer citation contained an article label

answer term coverage: fraction of expected answer terms found in the generated answer

refusal detected: whether the answer matched the defined refusal patterns

Operational Failure Thresholds

Historical automated metrics were reported as continuous means. Later author-reviewed pair-success endpoints are defined separately in S3 and S8. Historical operational flags identified rows requiring review:

Table S8 Evaluation audit flags

Audit condition	Flag used
Answerable row with no retrieved gold source	wrong_or_missing_source
Answerable row with article Recall@k below 1.0	incomplete_article_retrieval
Retrieved expected-term coverage below 0.50	low_evidence_term_coverage
Answerable row that returned the refusal sentence	incorrect_refusal
Answerable row without a recognized citation	missing_citation
Citation-like label not present among retrieved citation labels	unsupported_or_unretrieved_citation
Generated answer-term coverage below 0.50	incomplete_answer
AI-judge faithfulness below 0.80	judge_flagged_unfaithful
Unanswerable row without literal refusal detection	failed_to_refuse

The English completeness diagnostic experiment used AI-judge completeness below 0.80 to identify low-completeness answerable English rows. These thresholds were used for technical error analysis and targeted repair. They were not regulatory acceptance criteria.

Automated AI Judge

The automated judge used models/gemini-flash-lite-latest with temperature 0.0. It received the question, system answer, expected answer, answerability flag, and retrieved evidence, then returned four numeric scores from 0 to 1: faithfulness, citation correctness, completeness, and refusal correctness. The judge was used as a structured technical scoring tool, not as an independent regulatory expert panel. The answer model

and automated judge used the same unpinned Gemini alias. Historical resolved backend versions were not retained; the judge was not independent.

The exact judge prompt, scoring definitions, model settings, row-level scores, rationales, and stage-level summaries are provided only in Supplementary Data 2 to avoid duplication across supplementary files.

Confidence Intervals

When confidence intervals were reported, the method was pair-clustered percentile bootstrap:

Table S9 Historical bootstrap settings

Setting	Value
Confidence level	95%
Resamples	2000
Seed	20260619
Cluster key	bilingual pair id

Pair clustering was used because Portuguese and English questions were written as paired items.

Reproducibility Boundary

The package supports reproduction of the reported technical evaluation if the same code, corpus files, model identifiers, benchmark files, API model behavior, and package versions are available. A conda environment export and pip freeze file are included as environment.lock.yml and requirements-freeze.txt. Exact generated answers may still vary because hosted LLM behavior can change over time even when the model name and prompt are fixed. The archived per-question CSVs are therefore part of the evidence record and should be used to audit the reported results.

The corresponding author wrote the course notebooks and Shenia code used in the development history. The coauthor contributed regulatory framing, interpretation and review. The archived Streamlit smoke test answered English in Portuguese and is not evidence of bilingual deployment equivalence. Its old screenshot and interface schematic are omitted from the submission figures.

The original refusal strings, prompts and row-level records remain exact research data, including missing diacritics or wrong-language outputs. Editorial normalization does not alter generated responses. The implemented encoder uses CLS pooling in the archived wrapper; model-card mean pooling was tested separately. Historical backend versions are unresolved; later recorded responses identify gemini-3.5-flash-lite. Full chronology and prompt hashes are in the research archive.

S2 Assessment sizes overlap and statistical handling

Benchmark. The constructed main benchmark contained 200 question rows organized as 100 Portuguese, English pairs: 140 direct-article questions, 20 multi-article or exception questions, and 40 unanswerable or adversarial questions. Each language had 100 rows (80 answerable and 20 unanswerable). This construction total is distinct from the evaluated sample sizes. The initial internal evaluation used 150 rows, comprising 120 answerable and 30 unanswerable questions; its 75 English and 75 Portuguese

questions were reused in the respective repair stages. Across all six reported stages there were 404 outputs and 254 distinct question IDs, without implying semantic independence of those IDs. The audit recorded 78 unique gold article citations. Gold answers cited RDC 9/2015 in 28 of the 200 question rows (14 bilingual pairs), including 18 of the 150 initial evaluation rows. These counts identify exposure to a revoked source, not the number of answers adjudicated incorrect under current law. Figure 2 reports distinct question rows per source; multiple articles from the same regulation in one question count once. Benchmark structure and archived gold source coverage are shown in Figure 2; evaluated denominators are provided in Supplementary Section S2.

Evaluation stages and leakage control. Development runs were used to identify and repair failures. The initial internal evaluation comprised 150 questions (75 English and 75 Portuguese). Subsequent English and Portuguese repair stages reused all 75 questions in their respective language subsets, with identical question IDs and text. Once these questions informed repair, the initial set served as an internal development baseline rather than an independent held-out test. Scores across these stages describe paired, within-language changes on reused development questions and do not establish generalization. When internal results became high, an AI-authored diagnostic stress test was run once to test whether the internal pattern generalized. Findings from this stress test informed subsequent refusal-prompt development. The 24-row calibration set contained three substantively equivalent bilingual question pairs from the 50-row stress test (six rows in each set), covering the current SUSAR notification form, the current medical-device notification fee, and the patient-identification premise. Four additional bilingual pairs had related content but asked narrower or differently framed questions. Different IDs and wording therefore did not establish independence. The later 30-row refusal-focused assessment was conducted after calibration and is interpreted as a diagnostic assessment within the same development programme, not independent validation.

Statistical handling and audit trail. Source-level retrieval was summarized by source hit and source MRR. Article-level retrieval was summarized by Recall@k, Precision@k, MRR@k, and nDCG@k after lowercasing, accent removal, and whitespace normalization of citation labels. For generated answers, the system recorded citation presence, article citation presence, retrieved-citation validity, answer-term coverage, refusal detection, and AI-judge scores. When confidence intervals were reported in supplementary files, pair-clustered percentile bootstrap resampling used bilingual pair ID as the cluster key, 2000 resamples, 95% confidence level, and seed 20260619. For the revised abstract and Supplementary Section S2, confidence intervals were recalculated from archived question-level values with the same 2,000-resample method and seed. Both languages of a question pair were retained together; language-specific repair stages retained their evaluated language only. Intervals and denominators use the original answerability labels and remain conditional on those labels, the source snapshot and the automated judge. They do not incorporate uncertainty from label errors, outdated law or judge error. An all-identical set yields a degenerate percentile interval; this does not establish perfect population performance. No confidence interval for a between-set calibration effect is reported because the assessments differ and their development was not independent.

Table S10 Historical evaluation stage denominators

Stage	Outputs	Pair IDs	EN / PT	Answerable / unanswerable
S1 Internal development baseline	150	75	75 / 75	120 / 30
S2 English repair on reused questions	75	75	75 / 0	60 / 15
S3 Portuguese repair on reused questions	75	75	0 / 75	60 / 15
S4 Diagnostic stress assessment	50	25	25 / 25	40 / 10
S5 Refusal calibration development	24	12	12 / 12	8 / 16
S6 Post calibration diagnostic assessment	30	15	15 / 15	12 / 18

Table S11 Historical endpoint estimates and uncertainty

Stage and endpoint	Outputs / pairs	Estimate	95% CI
S4 Faithfulness	40 / 20	1.000	1.000 to 1.000
S4 Citation correctness	40 / 20	0.975	0.925 to 1.000
S4 Completeness	40 / 20	0.943	0.873 to 0.990
S4 Correct refusal	10 / 5	7/10 (0.700)	0.300 to 1.000
S6 Faithfulness	12 / 6	1.000	1.000 to 1.000
S6 Citation correctness	12 / 6	1.000	1.000 to 1.000
S6 Completeness	12 / 6	0.908	0.792 to 1.000
S6 Correct refusal	18 / 9	17/18 (0.944)	0.833 to 1.000

S3 Author assessments and answerability adjudication

On 26 October 2026, coauthor H.L.R.J. completed author verification of 94 selected historical outputs (93 distinct question IDs; 50 bilingual pair IDs). The selection included all 44 originally unanswerable outputs from the stress, calibration and post-calibration stages, 40 outputs sampled by originally answerable pair ID across the six stages, and ten originally unanswerable internal-baseline outputs. The returned ratings were AI-assisted: a Claude-prepared draft was personally reviewed and approved by the coauthor, as confirmed by the corresponding author. The coauthor had seen some automated scores, reference answers or coordinator information before approval; the planned blinded initial assessment was therefore not achieved. Ratings were frozen before this descriptive comparison. Citation support and completeness were compared with judge passes at the prespecified threshold of 0.80, with 1.00 as a sensitivity check; refusal comparisons were restricted to questions classified as unanswerable by the coauthor. Partly answerable and uncertain cases were reported separately. Exploratory 95% percentile intervals used 10,000 bilingual-pair bootstrap draws, retaining selected languages and stages within each pair. This author review is not independent validation; reconstructed whole-article evidence does not establish faithfulness to the exact original retrieved context. Following corresponding-author adjudication, refusal results were stratified by final archived-corpus answerability. For the small all-pass unanswerable

strata, two-sided exact 95% binomial intervals used a bilingual-pair endpoint requiring both language outputs to be appropriate refusals. These intervals assume independent question pairs within each stratum and do not account for review or adjudication bias; they are distinct from the earlier output-mean bootstrap intervals.

Frozen author-led assessment. Twenty bilingual pairs (40 questions: 12 answerable, four partly answerable and four unanswerable pairs) and their references, source scope and scoring plan were frozen before generation on 30 October 2026. The original six-document index and prompt produced one answer per question with gemini-3.5-flash-lite at temperature 0.2, without retries, repairs or new automated judging. Coauthor H.L.R.J. rated every answer against the exact selected passages and frozen reference. His confirmed review date was 30 October; the uploaded date of 29 October is preserved with a separate correction record. He reported no exposure to automated scores for this run. The primary endpoint required both languages of a pair to meet the prespecified category rules, including correct response language. Partly answerable questions additionally required supported information and explicit identification of the unavailable detail. Exact two-sided 95% binomial intervals were calculated at pair level. Unanswerable rows had no supported-content completeness endpoint; their raw Yes/Sufficient entries were treated as not applicable for derived completeness and retrieval summaries. The questions were AI-authored after earlier benchmark exposure and coauthor-approved, so this is reference-informed author-led assessment, not independent external validation.

The coauthor retained 40 originally answerable classifications and initially proposed changing six originally unanswerable rows to answerable and twelve to partly answerable or ambiguous; 36 remained unanswerable. The corresponding author approved these labels and reference drafts on 28 October 2026 in a separate overlay; the original benchmark remains preserved. Response appropriateness was rated yes for 78 of 94 selected outputs and no for 16; these targeted-sample counts do not estimate population accuracy. For completeness, the author and judge agreed on 38/58 applicable outputs (65.5%; exploratory 95% interval 48.2 to 80.9%) at the 0.80 judge threshold: 37 both passed, one both failed, 16 passed the judge but failed author review, and four showed the reverse. Restricting completeness to the 46 questions the author classified as answerable yielded 37/46 agreement (80.4%; interval 63.8 to 93.6%). These different denominators distinguish rating disagreement from answerability changes.

On 28 October 2026, the corresponding author approved the nine bilingual pair classifications and their draft reference answers after the archived-source audit: six formerly unanswerable outputs were classified answerable and twelve partly answerable, with the initial human ratings preserved. In the fully reviewed stress, calibration and post-calibration unanswerable strata, author-rated appropriate refusals were 4/4, 10/10 and 12/12 outputs, respectively (2/2, 5/5 and 6/6 bilingual pairs). Exact 95% binomial intervals for the endpoint that both language outputs in a pair were appropriate were 15.8 to 100.0%, 47.8 to 100.0% and 54.1 to 100.0%, respectively. These pair-level intervals are not independent output-level intervals and do not account for unblinded review, AI assistance or post hoc label decisions. Historical judge passes on the retained strata were 4/4, 9/10 and 12/12; these remain archived scores, not automated rescoring against the revised references. The 12 partly answerable outputs remain a separate

stratum. No between-stage improvement or independent validation follows from these selected, small and differently composed cohorts.

In the first-repeat author review, supported-information completeness was rated Yes for 16/18 answers without sibling expansion and 18/18 with expansion among changed questions. Selected evidence was rated sufficient in 11/18 and 17/18, respectively; language compliance was 15/18 and 16/18. Across all 44 unique answers, 41 were rated complete and 38 language-compliant. Three refusals appropriately withheld unavailable detail but omitted supported information and were rated incomplete; one was an unchanged counterpart. These descriptive author judgments remain susceptible to score and condition exposure and do not independently confirm the automated effect or current-law accuracy.

In the frozen author-led assessment, 10/20 bilingual pairs met the full task criteria (50.0%; exact 95% CI 27.2 to 72.8%). Category results were 6/12 answerable pairs (50.0%; 21.1 to 78.9%), 0/4 partly answerable pairs (0%; 0 to 60.2%) and 4/4 unanswerable pairs (100%; 39.8 to 100%). At output level, 24/40 passed: 16/24 answerable, 0/8 partly answerable and 8/8 unanswerable. Full reference content was covered in 18/24 answerable outputs, but English language failures reduced complete-task success. Requested language was respected in 35/40 outputs: 15/20 English and 20/20 Portuguese. Table 2 and Figure 3 report the paired outcomes; supplementary audit S3 provides all ratings and derived results.

The selected passages confirm missing list items in AV_004, AV_006 and AV_014, absence of the required Article 28 item in AV_013 Portuguese, and missing portions of the paired provisions in AV_016. AV_009 Portuguese correctly supplied the registration classes but could not supply the notification classes from its passages. All eight partly answerable outputs were blanket refusals. Five omitted information that Howard judged supported by the selected passages; three lacked the required supported content in those passages. None identified the unavailable detail explicitly. The frozen prompt required a blanket refusal for partial evidence, whereas the prespecified partly answerable endpoint required a supported partial answer. This instruction and endpoint mismatch is a documented design limitation. AV_008 English cited supported pharmacovigilance statements about risk minimization instead of answering the contingency-plan question. All 23 claim-bearing outputs had supported claims and citations according to Howard, illustrating why grounding alone is not task success. These failure categories overlap and do not identify causal effect sizes.

Table S12 Historical author review sample and labels

A = answerable; P = partly answerable; N = unanswerable. Selected outputs total 94 across stages; repeated question IDs across stages are retained. Changed labels are a separate adjudication overlay, not replacement historical judge scores.

Stage	Selected outputs	Originally A / N	Author A / P / N	Changed labels
S1 internal baseline	18	8 / 10	8 / 0 / 10	0
S2 English repair	4	4 / 0	4 / 0 / 0	0
S3 Portuguese repair	4	4 / 0	4 / 0 / 0	0
S4 stress assessment	18	8 / 10	12 / 2 / 4	6
S5 calibration development	24	8 / 16	10 / 4 / 10	6
S6 post calibration	26	8 / 18	8 / 6 / 12	6

Table S13 Historical author and judge agreement

Endpoint and threshold	Eligible	Both pass / fail	Author pass judge fail	Author fail judge pass	Agreement and 95% interval
citation support at 0.80	44	44 / 0	0	0	100.0% [100.0, 100.0]
completeness at 0.80	58	37 / 1	4	16	65.5% [48.2, 80.9]
refusal at 0.80	36	35 / 0	1	0	97.2% [90.9, 100.0]
citation support at 1.00	44	43 / 0	1	0	97.7% [91.9, 100.0]
completeness at 1.00	58	34 / 1	7	16	60.3% [43.4, 76.1]
refusal at 1.00	36	35 / 0	1	0	97.2% [90.9, 100.0]

S4 Archived sources and targeted legal status

Regulatory corpus. The corpus included six selected ANVISA documents covering clinical trials, biological products, medical devices, pharmacovigilance, and reliance-based review [1,2,6-9]. The selected documents concern common drug-development and health-product topics; this selection did not establish that every indexed provision was in force at evaluation. The final corpus included IN 338/2024, RDC 406/2020, RDC 55/2010, RDC 751/2022, RDC 945/2024, and RDC 9/2015. Corpus composition and use cases are summarized in Table 1. RDC 9/2015 was already revoked by Article 98 of RDC 945/2024 when these evaluations were conducted; Anvisa identifies 1 January 2025 as the latter regulation's effective date. Its presence in the archived corpus and gold answers is a benchmark limitation, not a deliberately designed temporal-validity test. Article 90 of RDC 945/2024 contains submission-date transitional provisions; applicability cannot be inferred for a benchmark question without the relevant dates [1,2].

Each source file was checked against its document identifier and topic before final indexing. Source verification identified one local-file metadata mismatch: the local file name suggested stability studies, while the document content was RDC 406/2020 on pharmacovigilance. The final corpus table therefore used official regulation numbers and topics rather than local file names. Identifier and topic checks did not establish consolidated legal validity. A subsequent status audit checked official texts on 23 October 2026 and identified 18 additional constructed-benchmark rows citing wholly revoked articles of RDC 55/2010, alongside the 28 rows citing revoked RDC 9/2015 (46 of 200 rows, or 23 bilingual pairs). The RDC 55/2010 revocations follow RDC 875/2024 and RDC 948/2024 [10,11]. Consolidated texts also show amendments under IN 345/2025, RDC 777/2023 and RDC 967/2025 for

IN 338/2024, RDC 751/2022 and RDC 406/2020, respectively. Supplementary Section S4 records the source checks and their limits; whole-article exposure counts do not certify the remaining gold answers.

The two repaired refusal failures remain failures against the explicitly archived questions. A targeted current-source check supports the staffing criterion but identifies the cited ruminant-documentation article as revoked. This check does not establish current-law answer accuracy or complete a corpus-wide legal review.

Exact evidence and archived outcomes

All eight outputs for FV_014 and FV_016 had the supported core requirement in their selected passages but returned blanket refusals. They omitted supported content and did not identify the unavailable headcount or form code. The repaired condition therefore retains two failed bilingual pairs. Context hashes and outputs are in the targeted-audit data. No generation or judge calls were repeated.

Provision status checked

FV_014: Consolidated RDC 406/2020 Article 18 retains the requirement for qualified staffing compatible with pharmacovigilance activities. It specifies no numerical minimum. This is not an exhaustive finding that no other applicable instrument could impose additional requirements.

FV_016: Consolidated RDC 55/2010 explicitly marks Article 22 revoked by RDC 948/2024. Article 59 II of RDC 948 lists this repeal; Article 60 sets commencement at 1 January 2025. The frozen Article 22 reference remains a historical reading target and cannot itself establish a current documentation obligation.

FV_012: The broad archived question requires Articles 17, 30 VIII-IX and 31 I-III. The corresponding author explicitly confirmed that scope on 1 November 2026. Article 30 is revoked by RDC 948/2024, while Article 17 and Article 31 I-III remain displayed in the consolidated text checked. Completed author review gives 13/20 versus 17/20 expanded-scope pairs; the original AI primary scores remain 14/20 versus 18/20. Scope confirmation does not establish current-law applicability. The later coauthor review covers all four generated answers separately and marks IRH_051 as failing citation support; the original scope-pending text in the uploaded draft is preserved as provenance rather than a reversal of the already recorded scope decision.

Separate transmissible spongiform encephalopathy rules and limits

AnvisaLegis lists RDC 305/2002 as in force with amendments. Consolidated RDC 68/2003 Article 1 addresses information for human-use products with ruminant-derived starting materials. Article 12 requires an analytical quality-control report supporting Q1 information and Q3 documents at registration or exemption requests, among other events. The Q1 form includes medicines. These separate provisions show why repeal of RDC 55 Article 22 must not be equated with elimination of transmissible spongiform encephalopathy controls (EET in the Portuguese sources). Coverage, material and geographical risks, exceptions, amendments and current filing rules need product-specific assessment. These additional instruments were outside the frozen corpus and are not imported retrospectively into its gold answers.

Interpretation

Current-law rebuilding and reevaluation and fresh independent validation remain incomplete or absent. All 80 earlier repair outputs have personal author review: 44 corresponding-author confirmations and 36 coauthor-reviewed AI-prepared ratings. The subsequent common-prompt comparison and BM25 extension also have completed coauthor assessment (S10); their AI and human ratings remain separate. Frozen references, answers and scores are preserved.

RDC 406 consolidated

RDC 55 consolidated

RDC 948 consolidated

The official consolidated texts were checked on 1 November 2026. Exact URLs and access scope are retained in official_sources.json in the targeted-audit data.

Exposure and screening categories do not identify adjudicated current-law error counts. Unapproved replacement drafts and authoring templates are omitted from the submission. The original source snapshots and score lineage remain preserved in the research archive.

S5 Retrieval ablations and embedding length

The archived index used different tokenizers for chunking and embedding. Thirty-five of 481 nodes exceeded the 512-WordPiece cap (maximum 866); full text remained available to BM25 and generation. Window pooling changed those 35 vectors while holding node identities and queries fixed. The recorded CLS wrapper reproduced the archived vectors; mean pooling was a separate diagnostic, not a multilingual-model comparison. Detailed token counts, all rule triggers and each regression remain in the data.

Retrieval-component comparisons used all 160 answerable development questions (80 bilingual pairs), the fixed 481-node historical index and archived gold citations. English retrieval queries used the saved offline glossary expansion; Portuguese queries were unchanged, and reranking used the original question. We compared BM25, dense retrieval, fusion, reranking without anchors, and sequential restoration of query anchors, article/lexical anchor rules and neighbor expansion. Raw top-five node results and citation-diversified top-five results were reported separately; like-for-like contrasts retained citation diversification. The implemented embedding wrapper used CLS pooling. A separate diagnostic applied the model card’s attention-mask-weighted mean pooling [20] with the same checkpoint and 512-token cap. Paired 95% intervals used 10,000 bilingual-pair bootstrap resamples; these exploratory intervals were not adjusted for multiple comparisons. No new answer-generation or judge calls were used for these ablations.

In the 160-question fixed-data retrieval comparison, citation-diversified BM25 article Recall@5 was 0.6969, the implemented CLS-pooled dense baseline 0.3344, and reciprocal-rank fusion 0.5687. Reranking without anchors or neighbor expansion yielded 0.9500; restoring query-anchor text yielded 0.9563, restoring article/lexical anchor rules yielded 0.9688, and the full pipeline yielded 0.9875. The paired reranker contribution relative to fusion was +0.3812 (95% interval +0.2969 to +0.4656); the neighbor-expansion contrast was +0.0187 (95% interval +0.0000 to +0.0437). With the full downstream rules retained, BM25-only candidates yielded 0.9875 and dense-only candidates 0.9094; adding dense candidates to the BM25 full pipeline changed recall by +0.0000 (95% interval -0.0187 to +0.0187). The equal mean for BM25 with full rules and the hybrid pipeline conceals one question gained and one lost when dense candidates were added; their paired interval spans zero. There is no demonstrated incremental mean Recall@5 benefit from the dense branch in this fixed comparison, without an equivalence claim. These are article-level retrieval comparisons, not answer-quality results.

The same-checkpoint mean-pooling diagnostic yielded dense Recall@5 0.3531 and fused Recall@5 0.6031; the dense contrast against implemented CLS pooling was +0.0187 (95% interval -0.0656 to +0.1000). The diagnostic changes the embedding representation, not language coverage. Supplementary Section S5 reports all conditions, language strata, paired gains and regressions. The results do not by themselves identify a preferred multilingual encoder or establish that the full pipeline is optimal.

Table S14 Retrieval ablation conditions

Condition	Recall@5 all	Recall@5 EN	Recall@5 PT	MRR@5 all
BM25 raw nodes	0.6906	0.4562	0.9250	0.5292
Dense CLS raw nodes	0.3281	0.2313	0.4250	0.1901
Fusion raw nodes	0.5563	0.3688	0.7438	0.3787
BM25 diversified	0.6969	0.4688	0.9250	0.5319
Dense CLS diversified	0.3344	0.2437	0.4250	0.1919
Fusion diversified	0.5687	0.3937	0.7438	0.3832
Rerank no anchors	0.9500	0.9375	0.9625	0.8634
Plus query anchors	0.9563	0.9500	0.9625	0.8692
Plus article/lexical anchors	0.9688	0.9500	0.9875	0.8723
Plus neighbors full pipeline	0.9875	0.9750	1.0000	0.8673
Dense mean pooling diversified	0.3531	0.3125	0.3937	0.2145
Fusion mean pooling diversified	0.6031	0.5188	0.6875	0.4000
BM25 then full rules	0.9875	0.9750	1.0000	0.8748
Dense CLS then full rules	0.9094	0.8688	0.9500	0.8036

Table S15 Paired retrieval ablation contrasts

Contrast	Recall difference	95% interval	Gains / losses
Dense minus BM25	-0.3625	-0.4656 to -0.2625	8 / 68
Fusion minus BM25	-0.1281	-0.2095 to -0.0469	10 / 31
Reranker minus fusion	+0.3812	+0.2969 to +0.4656	64 / 2
Add query anchors	+0.0063	+0.0000 to +0.0187	1 / 0
Add article anchors	+0.0125	+0.0000 to +0.0312	2 / 0
Add neighbors	+0.0187	+0.0000 to +0.0437	5 / 0
Mean minus CLS dense	+0.0187	-0.0656 to +0.1000	25 / 22
Mean minus CLS fusion	+0.0344	-0.0281 to +0.1000	15 / 10
Add dense branch	+0.0000	-0.0187 to +0.0187	1 / 1
Add BM25 branch	+0.0781	+0.0344 to +0.1281	13 / 1

S6 Answer effects of embedding coverage and sibling expansion

The targeted answer comparison used all 30 affected-gold questions (15 bilingual pairs) and 20 seeded gold-unaffected controls (10 pairs), with three generations per condition, giving 300 answers and 300 automated evaluations. The fixed randomized plan, original QA prompts, citation-repair policy and judge rubric were retained. Every successful call

returned modelVersion gemini-3.5-flash-lite; generation temperature was 0.2 and judge temperature 0. Scores were masked to condition labels. The primary endpoint was the full-coverage minus prefix mean automated completeness in the affected stratum, averaging three repeats per question before 10,000 bilingual-pair bootstrap resamples. Secondary and subgroup analyses were exploratory without multiplicity adjustment. Provider quota interruption required resumption on the following day; successful exact-prompt responses were reused, without regenerating completed calls. Intervals condition on the three observed repeats and the archived gold and do not estimate independent human agreement or isolate temporal provider variation.

Among the 18 changed questions, equal-pair-weighted automated completeness was 0.793 without sibling expansion and 0.969 with expansion, an observed increase of +0.176 (95% pair-bootstrap interval +0.008 to +0.364). Question-level repeat means improved in 8 questions, worsened in 1 and were unchanged in 9. This is an observed development-data contrast, not an independent estimate of generalization. The eight unchanged counterpart questions share exact input-response records across arms by design and are not independent stochastic-noise controls. Scores are provisional same-model assessments against archived references; all regressions, refusals, language flags, citation checks and failures are retained. Unblinded coauthor review of the 44 unique first-repeat answers was subsequently completed and is reported separately. A post hoc sensitivity excluded three pairs with previously flagged revoked gold (RDC 9/2015 or wholly revoked RDC 55/2010 articles), retaining 13 changed questions across 10 pairs. Automated completeness was 0.822 without expansion and 0.995 with expansion (difference +0.173; 95% interval +0.000 to +0.408). That interval reaches zero. This screen does not verify the remaining references or establish current-law accuracy. The prespecified archived-gold primary result is retained unchanged.

In the first-repeat author review, supported-information completeness was rated Yes for 16/18 answers without sibling expansion and 18/18 with expansion among changed questions. Selected evidence was rated sufficient in 11/18 and 17/18, respectively; language compliance was 15/18 and 16/18. Across all 44 unique answers, 41 were rated complete and 38 language-compliant. Three refusals appropriately withheld unavailable detail but omitted supported information and were rated incomplete; one was an unchanged counterpart. These descriptive author judgments remain susceptible to score and condition exposure and do not independently confirm the automated effect or current-law accuracy.

The judge’s refusal rubric uses whole-corpus answerability, whereas the generator is instructed to refuse when its retrieved context is insufficient. In the new 300-output pilot, 89 outputs were refusals and 50 of those had no archived gold citation label in context. Gold-label absence is a retrieval proxy, not an adjudication of passage sufficiency. A low completeness score measures failure to deliver the reference information; it does not establish unsafe refusal behavior. All 300 automated faithfulness scores were 1.000, including the 89 refusals, so that endpoint did not discriminate among these outputs. Human evidence-sufficiency and source-applicability review remains pending. English questions also received the exact Portuguese refusal sentence in 22 of 150 outputs (11 per condition), directly from generation rather than repair or wrapper fallback; this exact-string screen does not count every possible language mismatch.

In the embedding-coverage answer experiment, affected-stratum completeness was 0.5900 versus 0.6167 (difference +0.0267; 95% interval -0.0267 to +0.0989); controls were 0.7983 versus 0.8133 (+0.0150; -0.0150 to +0.0600). Only 10/50 final answer prompts changed. Across the 40 byte-identical prompts, mean scores still changed from 0.7117 to 0.7200 (+0.0083; -0.0175 to +0.0390). This and the retained source/judge limits preclude attributing all score differences to coverage. Human verification of these 300 outputs remains incomplete.

Table S16 Sibling context answer comparison

Automated scores on 18 changed questions across 13 bilingual pair IDs, with three repeats per condition. Off and On denote sibling expansion. Repeat means are paired by bilingual question pair; raw refusal is the detected refusal proportion. These estimates are separate from the 44 unique first-repeat answers reviewed by the coauthor. The eight unchanged counterpart questions are excluded from this changed-question contrast.

Endpoint	Off	On	On minus off	95% pair interval
completeness	0.793	0.969	+0.176	[+0.008, +0.364]
faithfulness	1.000	1.000	+0.000	[+0.000, +0.000]
citation correctness	0.987	0.987	+0.000	[+0.000, +0.000]
raw refusal	0.115	0.000	-0.115	[-0.308, +0.000]

S7 Failure mechanisms negative results and reference scope

In the exploratory prompt comparison, AI-assisted inspection found that all requested information actually present in the passages was supplied in 1/5 evidence-eligible partly answerable outputs under the original prompt and 5/5 under the revised prompt. Explicit identification of unavailable detail changed from 1/8 to 6/8 partly answerable outputs, and blanket refusals from 7/8 to 2/8. Full frozen-task success changed from 0/4 to 1/4 partly answerable pairs and from 3/10 to 4/10 pairs overall; answerable and unanswerable control success remained 1/4 and 2/2 pairs in both conditions. These are provisional AI-assisted development counts, not Howard ratings or revised primary scores. Missing passages persisted. The revised prompt produced a wrong-plan answer in AV_008 English, an unrequested API definition in AV_013 Portuguese, three blanket refusals with prohibited citations, and a Portuguese answer to AV_016 English where the original prompt answered in English. Requested-language compliance was 17/20 versus 18/20 overall, concealing that individual regression. No general repair, independent improvement or current-law validity is established.

The combined comparison completed all 240 generations in 240 successful attempts. Full-task bilingual-pair success was 8/20, 9/20 and 9/20 under the original condition, versus 14/20, 15/20 and 14/20 under the combined condition across the three repeats. Mean pair success across repeats changed from 43.3% to 71.7%, a paired difference of 28.3 percentage points (descriptive 95% pair-cluster bootstrap interval 10.0 to 48.3). Seven of 20 original-condition pairs and 14/20 combined-condition pairs passed all three repeats. Combined-condition answerable pairs passed 10/12 in every repeat, and unanswerable pairs 4/4; partly answerable pairs passed 0/4, 1/4 and 0/4, with none passing all repeats. Wrong-language outputs decreased from 16/120 to 3/120, but literal-citation format failures increased from 5/120 to 7/120. The original 120 and combined 120 outputs contain repeated responses to the same 40 questions and are not

independent samples. These provisional AI-assisted outcomes do not replace Howard’s original 10/20 assessment.

All 99 new answers completed in 107 HTTP attempts; eight 503 responses were retried successfully. In the fresh heading controls, acceptable outputs were 0/6 with the incorrect heading and no scope instruction, 4/6 with the incorrect heading plus instruction, and 6/6 in each corrected-heading condition. Across the two English language tasks, requested-language compliance was 3/6 with the original ending and 6/6 with either added instruction. The facilities control varied from 0/3 English in the earlier batch to 2/3 here, and the appeal control from 0/3 to 1/3, demonstrating variation for identical prompts. The three original partial-answer pairs supplied the main supported content in 18/18 outputs, identified all missing requested details in 17/18, and included the disputed secondary reference components in 0/18. Explicitly requesting those components produced them in 18/18, with 17/18 raw full-task passes and 18/18 after citation normalization. This changes the question and does not adjudicate the original golds. For the contingency question, Article 9 alone, first, last or in Portuguese-query contexts each produced correct answers in 3/3 outputs; contexts without it produced the wrong task in 3/3. In the original-prompt bridge, Article 9 again yielded 3/3 correct answers; without it, two outputs appropriately refused for insufficient selected evidence and one answered the wrong task. Original benchmark and Howard scores remain unchanged.

On 31 October 2026, the corresponding author reported human confirmation of the available diagnostic assessments. This confirmation was author-led, with the AI-assisted analyses visible; it is not an independent or blinded assessment and does not cover newly authored validation items or answers that have not yet been generated.

Reference-scope adjudication treated three secondary components as contextual rather than mandatory for the original questions: the complementary quality act in AV_013, the general low-risk eligibility frame in AV_014, and the notification certificate-copy rule in AV_016. The references retain these details as context. The required core remains the information actually requested, including explicit identification of unavailable details. This author-directed adjudication was carried out by the assistant and is distinguished from the preceding author confirmation. In a retrospective overlay on the 18 original-question outputs in S7, passes changed from 0/18 under the prior strict rubric to 17/18 under the scope-aligned rubric; one Portuguese answer still omitted identification of the unavailable indexing databases. These are three bilingual pairs repeated three times, not 18 independent questions. No answer was regenerated and no original score or Howard rating was replaced (S7).

Evidence controls held questions and saved prompts fixed, with three repeats per condition. The first four-question diagnostic used 39 unique calls for 45 condition records because unchanged inputs reused responses. Adding direct ethics or reliance provisions altered supported premise handling, while sibling completion alone did not guarantee the needed provision. The ethics follow-up crossed original/clarified instructions with original, direct-Article-8 and length-matched unrelated-Article-11 contexts; the clarified prompt with direct evidence succeeded across all three language/reconstruction groups, but length controls and wording changes were not separable proofs of one mechanism. Both Portuguese reconstructions and all failures are retained.

Automatic-context completion increased reference-fragment coverage from 20/32 to 26/32 and 31/32 positive rows, but raw full-task pair counts were 11/20, 10/20 and 12/20 across three repeats. Citation normalization yielded 13/20, 12/20 and 12/20; six wrong-language outputs remained. Portuguese-query reranking recovered the contingency answer in 3/3 versus 0/3 controls; an extra English cue added no observed benefit. A separate 12-call heading correction yielded 6/6 acceptable heading-only responses versus 0/6 earlier controls, with cross-batch limits. Fresh within-batch controls are summarized in the mechanism table below. These development contrasts do not establish generalization.

Table S17 Mechanism controls and negative results

Mechanism	Observed result	Limit
Article 41 heading	Incorrect: 0/6; incorrect plus scope: 4/6; correct: 6/6 with either instruction condition.	One bilingual pair, three repeats per language.
Language	Original ending: 3/6 English; either added ending: 6/6.	Two selected English tasks; identical controls vary across batches.
Question scope	Secondary components: 0/18 original questions; 18/18 explicit questions.	Broader questions change the task; original golds retained.
Article 9	Alone, first, last and Portuguese-query contexts: 3/3 correct each; absent: 0/3.	Manual insertion diagnoses availability; it is not retrieval validation.
Original prompt bridge	With Article 9: 3/3 correct; absent: 2/3 appropriate refusals and 1/3 wrong-task answer.	Six new answers; revised-prompt comparators from earlier batch.

Table S18 Archived reference scope decisions

Pair	Required in the original question	Retained as contextual information
AV_013	Seven API dossier categories; explicitly state that numerical quality limits are unavailable.	Article 29 defers quality requirements to a complementary act. The original question did not ask how that act will be issued.
AV_014	Robust meta-analysis and systematic-review evidence from indexed journals concerning safety and efficacy; explicitly identify the unavailable databases and minimum publication count.	General phase 3, known-safety and minimal-additional-risk frame remains relevant context. It is not a mandatory extra answer component for the literature-type question.
AV_016	Import/marketing only for devices manufactured within certificate validity; explicitly identify the unavailable specific-device list.	The notification petition certificate-copy requirement concerns a separate documentation task not asked in the original question.

The earlier diagnostic assessments received self-reported author confirmation on 31 October with AI-assisted analyses visible. This was not independent or blinded review. The context and prompt controls test selected known failures; response reuse, cross-batch variation and repeated answers limit causal inference. Negative findings include lack of a consistent embedding-coverage benefit, persistent partial-answer refusals, no added benefit of one English cue, and no observed Article 9 position penalty. New question wording changed reference scope and is not a repair of the original task.

S8 Exploratory integrated repair comparison

The frozen comparison completed 20 retrieval translations and 80 answers. In the initial AI assessment under the frozen rubric, the bundled repaired pipeline passed 18/20 bilingual pairs versus 14/20 for baseline. The difference was smaller with common citation formatting, and blanket refusal still prevented supported partial answers in two pairs. All 80 outputs received personal author review on 1 November 2026: 44 from the corresponding author in an unblinded review and 36 from H.L.R.J. after personally checking an AI-prepared rating draft. The corresponding author reports that H.L.R.J. had not seen separate AI scores, condition identities or previous assessment findings before finalizing those ratings. Draft assistance remains disclosed; this is coauthor assessment, not independent external validation. Completed author review yields 14/20 versus 17/20 pairs, with four gains and one loss (exact paired p = 0.375). This differs from the preserved initial AI result because IRH_051 fails specific citation support.

Frozen design and execution

The experimental pipeline was frozen before authoring questions and retained all historical assets. The 20-pair source-first test was frozen before answers. On 1 November 2026, we completed one answer per question and condition using gemini-3.5-flash-lite at temperature 0.2 in frozen randomized order. All 80 answers were inspected locally with AI assistance against frozen component references and exact selected passages. Primary pair success required both languages to satisfy content, evidence-boundary, language and citation-label criteria. Raw-output and common-formatter comparisons were retained; all 80 outputs subsequently received personal author review, with initial AI scores and later author decisions retained separately.

The baseline retained the original 481-node index, normalized CLS-pooled MiniLM embeddings (512-token limit), dense/BM25 top-100 fusion (RRF constant 60), cross-encoder, top-five selection, conditional article expansion, glossary and original prompt. The separate 482-node repair completes selected/adjacent articles by physical source occurrence. Structural conservation checks passed; reproduced historical embeddings had minimum cosine similarity 0.999999843. These checks do not establish current legal validity.

Questions and references were frozen before answers, after screening against 452 historical questions; one semantic duplicate was replaced. The 20 AI-authored bilingual pairs covered 12 answerable, four partly answerable and four unanswerable pairs across four named archived regulations. FV_011 and FV_019 share Article 53. Source-first selection, topic overlap and development exposure limit novelty and generalization.

Execution spanned 31 October to 1 November 2026 after a daily-quota interruption. Successful calls were reused; the 20 translations required 22 attempts and the 80 answers 80 attempts. No judge or repair calls were made.

Pair outcomes and uncertainty

Table S19 Exploratory pipeline and author review outcomes

Endpoint requiring both languages	Baseline	Integrated repair
Completed author review overall	14/20; 70.0% 95% CI 45.7 to 88.1%	17/20; 85.0% 95% CI 62.1 to 96.8%
Paired difference (95% bootstrap interval)	Reference	+15.0 points -5.0 to +35.0
Exact paired test	Reference	p = 0.375
Post hoc expanded-scope author review	13/20	17/20

Endpoint requiring both languages	Baseline	Integrated repair
Initial AI-assessed primary pipeline	14/20	18/20
Initial AI-assessed raw outputs	14/20	15/20
Initial AI-assessed common formatter	15/20	18/20
Initial AI-assessed content and boundary only	16/20	18/20

Primary pipeline scoring uses raw baseline output and the prespecified deterministic postprocessor on repaired output. The repair postprocessor maps only exact canonical refusals to the requested language and otherwise applies the existing citation-label sanitizer. Raw answers remain unchanged. Rules were prespecified before question authoring; the executable wrapper was materialized after input freeze but before any answer. No reference-specific transformation was added.

The initial AI-assessed primary paired difference was +20.0 percentage points (100000 pair-cluster bootstrap resamples; seed 20261101; percentile 95% interval +5.0 to +40.0). Four pairs improved and none worsened; the exact two-sided discordant-pair test gave p = 0.125. The bootstrap resamples only the observed discordance pattern, so its positive lower bound should not be treated as conclusive with four discordant pairs. Raw-output difference was +5.0 points (interval -20.0 to +30.0; p = 1.000). With the same citation formatter on both conditions, the difference was +15.0 points (interval 0 to +30.0; p = 0.250). These are limited, provisional comparisons, not a demonstrated population or independent-validation benefit.

Table S20 Category outcomes by assessment layer

Assessment and category	Baseline pairs	Repair pairs
Author review: Answerable	10/12	11/12
Author review: Partly answerable	0/4	2/4
Author review: Unanswerable	4/4	4/4
Initial AI assessment: Answerable	10/12	12/12
Initial AI assessment: Partly answerable	0/4	2/4
Initial AI assessment: Unanswerable	4/4	4/4

At language-row level, initial AI-assessed primary success was 31/40 versus 36/40; the rows are dependent within bilingual pairs and are descriptive counts. Wrong-language outputs were 2/40 versus 0/40. Seven baseline and four repaired positive outputs were blanket refusals. No unsupported factual value was identified in the local inspection; this does not certify absence of hallucination or safety outside these questions. All eight unanswerable language rows refused in each condition; the pair-level 4/4 exact interval is 39.8 to 100%. The completed author overlay passes 31/40 baseline and 35/40 repaired outputs; the expanded-scope author overlay passes 29/40 and 34/40. These are separate rating layers.

Evidence coverage and retained failures

All 32 answerable or partly answerable language rows had every frozen reference article label selected and full normalized reference word five-gram coverage in both conditions. This is a source-selection and lexical check, not an answer-quality metric. Mean selected context length was 5736.5 characters for baseline and 16487.5 for repair. No retrieval-coverage gain was observable at this ceiling, and increased context length confounds attribution of any generator effect to a single repair component.

The repaired condition still refused both languages of FV_014, whose selected Article 18 states that qualified staffing must be compatible with pharmacovigilance activities, and FV_016, whose selected Article 22 requires documentation controlling transmissible spongiform encephalopathy for ruminant-

derived substances. Neither answer supplied the supported criterion or documentation requirement, nor identified the specific unavailable headcount or form code. These four failures establish over-refusal with complete required evidence. The prompt has both a partial-answer rule and a false-premise refusal rule, plus an exact-subject check; interaction is a plausible untested explanation. These frozen outputs were retained without further tuning within this repair comparison; the later common-prompt ablation is a separate experiment (S10).

Four repaired non-refusal outputs had raw citation-label violations: both languages of FV_008, Portuguese FV_004 and English FV_012. Deterministic processing corrected the suffixes or combined bracket. One baseline output, Portuguese FV_009, had suffix citations; common formatting corrects it too. Raw-format failures are retained separately from substantive content, and the raw/raw comparison does not establish improvement.

Scope and dependence sensitivity

A post hoc reference-scope audit established that FV_012's broad manufacturer/site and GMP question requires Articles 30 VIII-IX and 31 I-III as well as Article 17 of archived RDC 55/2010. These additional requirements cover cell-bank manufacturer/storage and batch-certificate issuer details, together with Anvisa and manufacturing-country authority GMP certificate copies. Only the repaired English answer supplied the expanded scope. Applying this scope to the initial AI-scored layer reduced bilingual success to 13/20 baseline and 17/20 repaired pairs, retaining a four-pair difference. This is an output-informed sensitivity analysis; the frozen references and primary 14/20 versus 18/20 results remain unchanged. Excluding the entire disputed pair yields 13/19 versus 17/19 under the primary endpoint; the previously reported exclusion sensitivity is retained separately. The corresponding author confirmed this archived scope on 1 November 2026. Article 30 is revoked for current-law purposes; see S4. Coauthor review judged IRH_051 complete in expanded content but failed its citation support for the technical-report requirement. Under completed author review, the expanded-scope pair result remains 13/20 versus 17/20, with four gains and no losses (exact p = 0.125).

In the initial AI-assessed sensitivity, grouping FV_011 and FV_019 together produces 19 source-article clusters: primary difference interval +4.8 to +40.0 points, common-formatter interval 0 to +31.6. Other source and topic dependence remains across only four regulations. Exact success intervals treat pairs as units and remain conditional on the frozen tasks and provisional rubric. One generated answer per condition does not characterize repeated-run stability.

Interpretation and stopping decision

The bundled pipeline showed a favorable descriptive result with a material formatting contribution. The staffing and ruminant-documentation refusals remain failures. The corresponding author personally confirmed 44 outputs and the coauthor approved the remaining 36 AI-draft-assisted ratings on 1 November 2026. Separate scores, conditions and prior findings were reportedly unseen for the latter review. Independent external validation, current-law verification and reevaluation, and an official-site answer-quality comparator remain absent. A separate subsequent common-prompt model-only baseline is reported in S10 and does not change these frozen author-reviewed results. Further pipeline changes would consume this test as development data. This bounded repair comparison was closed without retuning its inputs or replacing its negative results. Subsequent common-prompt and BM25 comparisons were registered separately (S10). S4 separates the retained current staffing criterion from revoked Article 22 and Article 30 references. The completed author review estimates a 15.0-point difference (95% pair-bootstrap interval -5.0 to +35.0; exact p = 0.375), not a demonstrated improvement. A prepared AI-draft review is not independently elicited human scoring, even when condition identities and separate scores were reportedly unseen.

Reproducibility records

The archive preserves frozen inputs, screening, prompt hashes, translations, all raw and processed answers, component decisions, pair tables, intervals, sensitivity analyses and

code. Full corpora, vector assets, credentials and private review files remain local. Original scores and Howard files are preserved.

S9 Data provenance and reader guide

The data archive preserves original outputs, separate adjudications, review provenance and negative results. Its README maps files to the paper. Credentials, private coordination files and superseded drafts are excluded.

S10 Common prompt answer ablation and matched lexical retrieval baseline

These post hoc analyses reuse 40 development-exposed FV questions and saved archived contexts, separately from the earlier repair experiment. Generation used the same Gemini model, temperature 0.2 and neutral common instructions, with hybrid passages, BM25 passages or no passages; no web search, judge API or repair calls were used. Expanded FV_012 scope was frozen before generation. Initial Codex assessment was anonymized with limited masking; the mapping was technically accessible. The BM25 AI assessment was unmasked. H.L.R.J. personally drafted and approved all 118 distinct ratings on 2 November 2026, covering 120 condition records. He reported no AI-score, condition-key or prior-result exposure for these outputs, with earlier project/question exposure disclosed. Answer style could reveal conditions. Human ratings use the common page references, not selected generator contexts. This is coauthor assessment, not independent external validation. Appropriate abstention passes absent-detail items; citations are secondary.

Table S21 Coauthor assessed answer outcomes by topic and archived support

Subgroup	Hybrid passages	BM25 passages	Model only
Clinical trials	11/12	12/12	2/12
Pharmacovigilance	12/12	12/12	2/12
Medical devices	10/10	10/10	3/10
Biological products	6/6	4/6	2/6
answerable	23/24	23/24	0/24
partly answerable	8/8	7/8	1/8
unanswerable	8/8	8/8	8/8

Subgroups partition the same 40 language-specific questions per condition and are descriptive, not independent tests. Unanswerable denotes a requested detail absent from the archived reference, not necessarily unavailable online. Appropriate abstention passes. Initial AI component data contain 110 nested reference units across the same 40 questions: 0/110 versus 100/110 were missing for hybrid versus model-only. These remain historical AI counts; Howard supplied binary component judgments, not a new 110-component recount. Units are not extra questions or measured correction actions.

Author-rated successes were 39/40 hybrid, 38/40 BM25 and 9/40 model-only answers, or 19/20, 18/20 and 4/20 pairs. Hybrid minus model-only pair success was +75 points (descriptive 95% pair-bootstrap interval +55 to +90; exact p = 0.000061). Hybrid minus BM25 was +5 points (-10 to +20; exact p = 1.000), with two gains and one loss. All conditions appropriately abstained on eight absent-detail answers. The retained hybrid error equates DPI with an Investigator's Brochure; BM25 fails Portuguese FV_012 and English FV_016. Howard and AI primary labels agree on 117/118 distinct answers. Howard accepts HCR_056 / FV_015 English for model-only content but flags correction; citations are secondary, so these judgments are not contradictory.

Table S22 Matched BM25 and frozen hybrid retrieval coverage

Metric	BM25	Frozen hybrid
Mean gold-article Recall at 5	95.8%	100%
At least one required article found at 5	31/32	32/32
All required articles represented at 5	30/32	32/32
Full text of all required articles in five chunks	29/32	30/32
Mean reciprocal rank within top five	0.932	0.964
Full required article text after identical completion	31/32	32/32
Median returned tokens after completion	3697	3953.5

Both methods use the same 482 chunks, saved Portuguese queries for English questions, named-regulation filtering, five seed chunks and identical article-completion policy. There are 32 positive-target questions in 16 bilingual pairs; the eight absent-detail items have no positive gold target and are excluded. Median expanded context size is 3697 versus 3953.5 tokens; equal seed counts do not imply equal final context length. Gold targets distinguish physical article occurrences and the expanded FV_012 scope uses Articles 17, 30 and 31. Coverage is structural, not semantic sufficiency or generated-answer correctness.

Hybrid minus BM25 mean article recall was +4.2 percentage points (descriptive 95% whole-pair bootstrap interval 0 to +11.5); all-article presence at five differed by +6.25 points (0 to +15.625), full text after completion by +3.125 points (0 to +9.375), and MRR@5 by +0.031 (-0.057 to +0.130). Bootstrap resampling used 16 bilingual pairs, 20,000 draws and seed 20261102. Small paired differences do not establish reliable superiority or equivalence. BM25 missed one FV_012 Portuguese article at five that completion recovered, and FV_016 English remained unrecovered. Original-English BM25 all-article retrieval was 5/16 versus 15/16 with saved Portuguese translations; this secondary language sensitivity must not be mistaken for the primary matched hybrid contrast.

The separately frozen BM25 extension used the same common prompt and 40 questions. One exact-prompt success was reused; 39 new calls succeeded on their first attempts. Saved hybrid answers were comparators, with sequential timing and one generation per item limiting inference. All outputs now have personal coauthor assessment, without transmission of ratings or references to Google. Howard confirms the added intermediate-manufacturer name/address requirement in FV_012 Portuguese as unsupported by the explicit Articles 17/31 II list. Intermediate products do occur elsewhere in Article 31 paragraph 6; this does not place them in that explicit list. The earlier AI sensitivity accepting the addition remains separate and is not the author primary result. English FV_016 omits Article 22 content; its saved context lacks that article. Howard calls this a refusal against the common reference, which does not establish refusal despite received evidence. Original notes and disagreements remain intact.

Supplementary Figure S1 maps all 20 bilingual tasks and six answer outcomes using the completed coauthor content criteria. It adds no independent observations. Full answers, references, separate AI and author ratings, provenance clarifications and disagreements accompany the data. Howard rates hybrid citation support correct in 39/39 cited answers versus the AI 38/39, despite both flagging the DPI substitution as a content error. His BM25 citation count is 38/40 versus AI 39/40 because the English FV_016 article list is treated differently; correct support is 37/38 versus 38/39. Model-only material claim flags are 14/40 versus AI 13/40, and omissions 31/40 versus AI 32/40. These secondary judgments are not blended. The original returned provenance has truncated/contradictory fields; the corresponding author confirmed personal approval, date and self-drafting separately. Private condition keys and full source/index assets are excluded. Current-law accuracy and fresh independent validation remain unestablished; workflow benefit was outside the experiment.